

SEMESTER II

Course Code	Course Name	L	T	P	C
CSEG1043	Data Structures and Algorithms	4	0	1	5
Total Units to be Covered: 6		Total Contact Hours: 90			
Prerequisite(s):	Programming in C-CSEG 1025	Syllabus version: 1.0			

Course Objectives

1. Provide a clear understanding of the importance of data structures in organizing and manipulating data efficiently.
2. To introduce students to fundamental data structures and the properties, memory management, and basic operations of each data structure.
3. To offer practical experience in implementing data structures, common sorting, and searching algorithms.
4. Emphasize the use of data structures as tools for algorithmic problem-solving and apply their knowledge of data structures to solve real-world problems.

Course Outcomes

Upon completion of the course, the students will be able to

- CO 1.** State the significance and properties of the fundamental data structures.
- CO 2.** Implement common data structures while ensuring proper memory management and error handling.
- CO 3.** Illustrate expertise in understanding the common sorting and searching techniques with their complexities and implement them.
- CO 4.** Analyse real-world problems by understanding the trade-offs involved in identifying the appropriate data structure(s) based on problem requirements and using them to solve the problems efficiently.

CO-PO Mapping

Program Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
Course Outcomes															

CO 1	1	-	2	1	-	-	-	-	-	-	-	1	-	-	-
CO 2	1	1	2	1	-	-	-	-	-	-	-	-	-	-	-
CO 3	1	1	2	1	-	-	-	-	-	-	-	2	-	-	-
CO 4	1	2	2	2	-	-	-	-	-	-	-	2	-	-	-
Average	1	1	2	1.25	-	-	-	-	-	-	-	1.25	-	-	-

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“_” means there is no correlation

Syllabus

Unit I: INTRODUCTION

12 Lecture Hours

Overview, classification, and Importance of data structures in Programming/problem-solving, Basic terminology, and concepts: elements, operations, storage, memory allocation, garbage collection, and compaction. Iterative & Recursive approaches. Basic analysis of algorithms: Amortized analysis and Asymptotic Analysis. Array: Memory representation (1D and 2D), Array operations: insertion, deletion, searching. Applications of Array. Structure: Nested Structure, Function pointer as member of a structure, Self-referential structure. Anonymous Unions, ADTs.

Unit II: LINKED LIST

10 Lecture Hours

Singly-Linked List, Doubly-Linked List, Circular Linked List, Header List, and its operations. Sentinel node. Generalized Linked List. Skip List. Applications of Linked Lists: polynomial manipulation, implementation of other data structures.

Unit III: STACK & QUEUE

10 Lecture Hours

Stack data structure and operations. Queue data structure and operations. Implementation of Stack and Queue using Array and Linked List. Circular Queue. Deque and its types. Priority Queue. Applications: Stacks (Conversion of Infix to

Prefix/Postfix, Expression evaluation, a note on DFS in graph), Queues (Job scheduling, a note on BFS in graph).

Unit IV: TREE

10 Lecture Hours

Introduction to Tree data structure and its terminologies, Binary Tree: properties, traversal algorithms (level-order, in-order, pre-order, post-order). Threaded Binary Tree. Binary Search Trees (BST): properties, operations (insertion, deletion, searching). Balanced BSTs. AVL Tree: properties, rotations, operations (insertion, deletion). Red-Black Tree. Multi-way search Tree: properties. B-Tree: properties, operations (search, insertion, and deletion). Applications of AVL Tree and B-Tree. Binary Heaps: properties, heapify operations, Heap sorting.

Unit V: HASH TABLE & GRAPH

10 Lecture Hours

Hashing and hash functions. Hash table data structure: structure, collisions, collision resolution techniques, maintaining load factor. Applications of hash tables: dictionaries, symbol tables.

Introduction to graph data structure and its terminologies. Graph representations: adjacency matrix, adjacency list. Graph traversal algorithms: depth-first search, breadth-first search (BFS). Connected Components. Minimum spanning tree. Shortest path.

Unit VI: SORTING & SEARCHING

8 Lecture Hours

Stability and In-place properties, Internal and external sorting. Simple comparison-based sorting algorithms: bubble sort, selection sort, insertion sort. Lower bound for comparison-based sorting algorithms. Recursive implementation of merge sort, quicksort, and binary search. Complexities of common sorting and searching algorithms.

Total lecture Hours 60

References*

Textbooks	1. S. Lipschutz, "Data Structures with C", Schaum's Outline Series, McGraw-Hill Education (India) Pvt. Limited, 2017. 2. Y. P. Kanetkar, "Data structures through C", 4rd Edition, New Delhi: BPB, 2022.
Reference books	1. A. V. Aho, J. E. Hopcroft, and J. D. Ullman, "Data Structures and Algorithms", New Delhi: Pearson Education, 2003. 2. E. Horowitz, and S. Sahni, "Fundamentals of Data Structures in C", 2nd Edition, Hyderabad: University Press, 2008.
Web Resources	
Journals	
MOOCs, online courses	

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100

Data Structures and Algorithms Lab

List of Experiments

Experiment 1: Basic Data Structure

To demonstrate the use of array, structure, and union along with dynamic memory allocation.

Experiment 2: Link List Data Structure and its Applications

To experiment with the concept of pointers, structure, and dynamic memory allocation to realize linked lists, their types, and application.

Experiment 3: Stack Data Structure

To use arrays and linked lists to implement Stack and its applications.

Experiment 4: Queue Data Structure

To demonstrate the use of arrays and linked lists to implement different variants of Queue and its applications.

Experiment 5: Trees

To demonstrate the creation of a binary tree and working with tree traversal.

Experiment 6: Heaps

To create a heap data structure and implement its operations, and its applications.

Experiment 7: Hash Tables

To implement a hash table using various collision resolution techniques, and its applications.

Experiment 8: Graphs

To demonstrate the creation of graphs and working with graph traversal algorithms.

Experiment 9: Sorting algorithms

To implement common sorting algorithms.

Experiment 10: Searching algorithms

To implement various search algorithms on data structures.

Total Lab hours 30

References*

Textbooks	1. S. Lipschutz, "Data Structures with C", Schaum's Outline Series, McGraw-Hill Education (India) Pvt. Limited, 2017. 2. Y. P. Kanetkar, "Data structures through C", 4rd Edition, New Delhi: BPB, 2022.
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Web Resources	
Journals	
MOOCs, online courses	

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme: Continuous Assessment

Components	Quiz & Viva	Performance & Lab Report
Weightage (%)	50	50